Algebraik ifodalarni ko'paytuvchilarga ajratish

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

---

1. **Ko'paytuvchilarga ajratish nima?**

A) O'xshash hadlarni ixchamlash
B) Ifodani ko'paytuvchilar ko'paytmasi ko'rinishida yozish
C) Qavslarni ochish
D) Ifodaning qiymatini topish

---

2. **Ko'paytuvchilarga ajratish qaysi amalga teskari?**

A) Ildiz chiqarishga
B) Darajaga ko'tarishga
C) Qavslarni ochishga
D) Qo'shishga

---

3. **Ko'paytuvchilarga ajrating:** $3x + 15$

A) $3(3x + 5)$
B) $3(x + 5)$
C) $5(3x + 1)$
D) $3(x + 15)$

---

4. **Ko'paytuvchilarga ajrating:** $10a - 4$

A) $10(a - 4)$
B) $2(5a + 2)$
C) $2(5a - 4)$
D) $2(5a - 2)$

---

5. **Ko'paytuvchilarga ajrating:** $7m + 7n$

A) $7(m + n)$
B) $7(m + 7n)$
C) $7mn$
D) $m(7 + n)$

---

6. $12x$ **va** $18y$ **hadlarning umumiy ko'paytuvchisi (eng kattasi) qaysi?**

A) 6
B) 2
C) 3
D) 36

---

7. $8ab + 12ac$ **ifodaning hadlari umumiy ko'paytuvchisi (eng kattasi) qaysi?**

A) $4a$
B) 4
C) $8a$
D) $2a$

---

8. $5xy - 15x$ **ni ko'paytuvchilarga to'liq ajrating.**

A) $5x(y - 3)$
B) $5(xy - 3x)$
C) $5x(y - 15)$
D) $x(5y - 15)$

---

9. $9p + 3$ **ni ko'paytuvchilarga ajrating.**

A) $3 \cdot 3p$
B) $9(p + 3)$
C) $3(3p + 1)$
D) $3(3p + 3)$

---

10. **Qaysi tenglik to'g'ri?**

A) $6x + 8 = 2(3x - 4)$
B) $6x + 8 = 3(2x + 4)$
C) $6x + 8 = 2(3x + 8)$
D) $6x + 8 = 2(3x + 4)$

---

11. **Ko'paytuvchilarga ajrating:** $2a(x+y)+5b(x+y)$

A) $(x+y)(2a\cdot 5b)$ **B)** $(x+y)(2a+5b)$

C) $10ab(x+y)$ **D)** $(x+y)(2a-5b)$

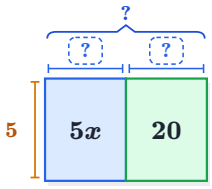
---

12. **Kalkulyatorsiz hisoblang:** $23\cdot 45+23\cdot 55$

A) $1\,035$ **B)** $2\,300$ **C)** 230 **D)** $2\,323$

---

13. **Rasmdagi to'g'ri to'rtburchakning yuzi $5x+20$, bo'yi 5. Enini toping.**

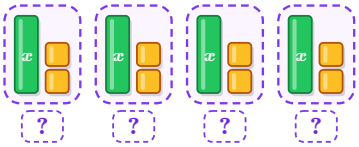


A) $x+4$ **B)** $x+5$

C) $x+20$ **D)** $5x+4$

---

14. **Rasmda $4x+8$ ta plitka 4 ta teng guruhga ajratilgan. Har bir guruhda nima bor?**

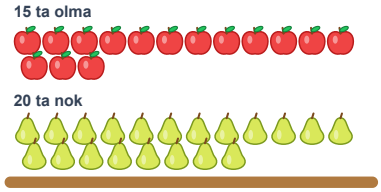


A) $x+2$ **B)** $4x+2$

C) $2x+4$ **D)** $x+8$

---

15. **Rasmdagi hamma mevalar 5 ta bir xil savatga joylandi, birorta meva ortib qolmadi. Har bir savatda nechta olma va nechta nok bor?**



A) 4 ta olma, 3 ta nok

B) 3 ta olma, 5 ta nok

C) 5 ta olma, 5 ta nok

D) 3 ta olma, 4 ta nok

---

16. **Ko'paytuvchilarga ajrating:** $-4a-8$

A) $-4(-a+2)$ **B)** $-4(a+2)$

C) $4(a+2)$ **D)** $-4(a-2)$

---

17. **Ko'paytuvchilarga ajrating:** x^2+5x

A) $x^2(1+5x)$ **B)** $5x(x+1)$

C) $x(x+5)$ **D)** $x(x^2+5)$

---

18. **Qaysi ajratish tugallangan?**

A) $12ma+6m=3(4ma+2m)$ **B)** $12ma+6m=6(2ma+m)$

C) $12ma+6m=6m(2a+1)$ **D)** $12ma+6m=2m(6a+3)$

---

19. **Ko'paytuvchilarga ajrating:** $14t+7$

A) $14(t+7)$ **B)** $7(2t+7)$ **C)** $7(2t+1)$ **D)** $7\cdot 2t$

---

20. **Ali $15a-10=5(3a-2)$ deb yozdi. Qavslarni ochib tekshiring: $5(3a-2)$ nimaga teng?**

A) $15a-10$ **B)** $15a+10$ **C)** $15a-2$ **D)** $8a-7$

Javoblar

Algebraik ifodalarni ko'paytuvchilarga ajratish · 1-test (oson)

№	Javob	Qiymati
1	**B**	Ifodani ko'paytuvchilar ko'paytmasi ko'rinishida yozish
2	**C**	Qavslarni ochishga
3	**B**	$3(x+5)$
4	**D**	$2(5a-2)$
5	**A**	$7(m+n)$
6	**A**	6
7	**A**	$4a$
8	**A**	$5x(y-3)$
9	**C**	$3(3p+1)$
10	**D**	$6x+8=2(3x+4)$

№	Javob	Qiymati
11	**B**	$(x+y)(2a+5b)$
12	**B**	$2\,300$
13	**A**	$x+4$
14	**A**	$x+2$
15	**D**	3 ta olma, 4 ta nok
16	**B**	$-4(a+2)$
17	**C**	$x(x+5)$
18	**C**	$12ma+6m=6m(2a+1)$
19	**C**	$7(2t+1)$
20	**A**	$15a-10$

Qisqa yechimlar

1. Algebraik ifodani ko'paytuvchilarining ko'paytmasi ko'rinishida yozish — ko'paytuvchilarga ajratish.
Javob: **ifodani ko'paytma ko'rinishida yozish**.

---

2. Qavslarni ochish: $3(x+2)=3x+6$. Ko'paytuvchilarga ajratish: $3x+6=3(x+2)$ — teskari yo'l.
Javob: **qavslarni ochishga**.

---

3. Umumiy ko'paytuvchi — 3: $3x=3\cdot x$, $15=3\cdot 5$.
$3x+15=3(x+5)$. Javob: $3(x+5)$.

---

4. EKUB$(10;\ 4)=2$: $10a=2\cdot 5a$, $4=2\cdot 2$.
Javob: $2(5a-2)$.

---

5. Ikkala hadda 7 bor: $7m+7n=7(m+n)$.
Tekshiruv: $7(m+n)=7m+7n$ ✓. Javob: $7(m+n)$.

---

6. Harflar har xil, demak faqat son: EKUB$(12;\ 18)=6$.
Javob: **6**. (3 va 2 ham umumiy, lekin eng kattasi emas.)

---

7. EKUB$(8;\ 12)=4$, ikkala hadda a bor (b va c — faqat bittasida).
Javob: $4a$: $8ab+12ac=4a(2b+3c)$.

---

8. Umumiy ko'paytuvchi $5x$: $5xy=5x\cdot y$, $15x=5x\cdot 3$.
$5(xy-3x)$ — qavsda hali x qolgan. Javob: $5x(y-3)$.

---

9. $9p=3\cdot 3p$, $3=3\cdot 1$ — had butunlay chiqsa, o'rnida 1 qoladi.
Javob: $3(3p+1)$.

---

10. Qavsni ochib tekshiramiz: $2(3x+4)=6x+8$ ✓. $2(3x+8)=6x+16$, $3(2x+4)=6x+12$.
Javob: $6x+8=2(3x+4)$.

---

11. Umumiy ko'paytuvchi — qavs $(x+y)$. Uni chiqaramiz, qavsda $2a$ va $5b$ qoladi.
Javob: $(x+y)(2a+5b)$.

---

12. 23 ni chiqaramiz: $23 \cdot (45 + 55) = 23 \cdot 100$.
Javob: **2 300**.

---

13. $5x + 20 = 5(x + 4)$. Yuz = bo'yi × eni, bo'yi 5.
Javob: $x + 4$.

---

14. 4 ta x plitka — har guruhga 1 tadan, 8 ta kvadratcha — har guruhga 2 tadan.
$4x + 8 = 4(x + 2)$. Javob: $x + 2$.

---

15. Rasmda 15 ta olma va 20 ta nok.
$15 : 5 = 3$ olma, $20 : 5 = 4$ nok. Ya'ni $15a + 20n = 5(3a + 4n)$. Javob: **3 ta olma, 4 ta nok**.

---

16. Minus bilan chiqaramiz: $-4a = -4 \cdot a$, $-8 = -4 \cdot 2$.
Tekshiruv: $-4(a + 2) = -4a - 8$ ✓. Javob: $-4(a + 2)$.

---

17. $x^2 = x \cdot x$, $5x = x \cdot 5$ — umumiy ko'paytuvchi x.
Javob: $x(x + 5)$.

---

18. Hammasi teng, lekin qavsda umumiy ko'paytuvchi qolmasligi kerak. $2ma + m$ da m, $6a + 3$ da 3 qolgan.
Javob: $6m(2a + 1)$.

---

19. $14t = 7 \cdot 2t$, $7 = 7 \cdot 1$.
Javob: $7(2t + 1)$.

---

20. $5 \cdot 3a = 15a$, $5 \cdot (-2) = -10$.
$5(3a - 2) = 15a - 10$ — Alining ajratishi to'g'ri. Javob: $15a - 10$.
